A Bayesian Network-Based Framework for Causal Cancer Drug Target Discovery Integrating Patient and Cell Line Data

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AT A GLANCE

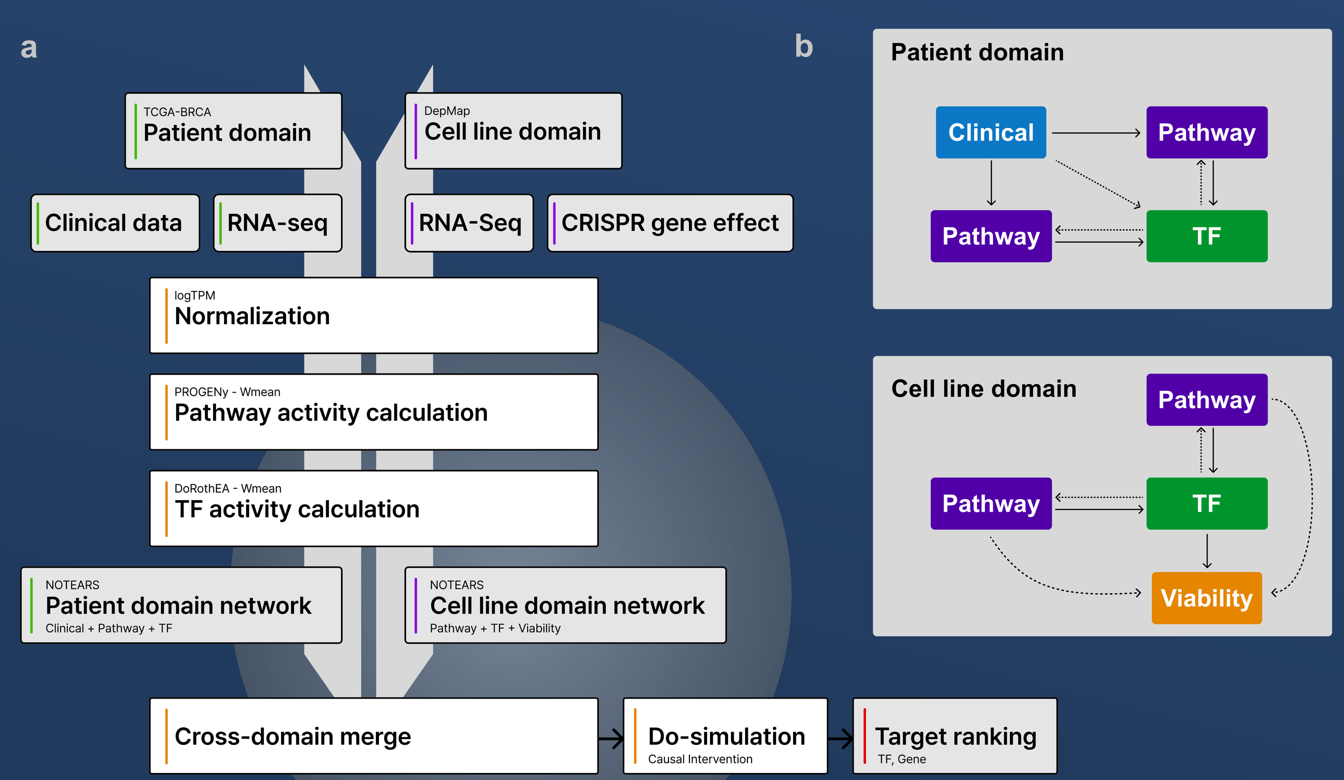
BayesTx framework predicts causal TF targets for any cancer using patient and cell-line consensus Bayesian Network

1,111 Patient cohort

51 Cell line data

275 Consensus edges

Overall Pipeline


a
TCGA-BRCA
Patient domain
DepMap
Cell line domain
Clinical data
RNA-seq
RNA-Seq
CRISPR gene effect
logTPM
Normalization
PROGENy - Wmean
Pathway activity calculation
DoRothEA - Wmean
TF activity calculation
NOTEARS
Patient domain network
Clinical + Pathway + TF
NOTEARS
Cell line domain network
Pathway + TF + Viability
Cross-domain merge
Do-simulation
Causal Intervention
Target ranking
TF, Gene
b
Patient domain
Clinical
Pathway
Pathway
TF
Cell line domain
Pathway
Pathway
TF
Viability

- **Patient dataset**: 1,111 patients in TCGA-BRCA, RNAseq + clinical.
- **Cell line dataset**: 51 BRCA cell lines in DepMap, CCLE2019 RNAseq + CRISPR dependency.
- **Pathway and Transcription Factor (TF) activiy calculation** DoRothEA (TF, regulatory network), PROGENy (Signaling Pathway).
- **BN structure learning:** NOTEARS on each domain
- **Cross-domain merge:** Build consensus network from two networks
- **TF target ranking:** Rank nodes with do-simulation scores on Viability.

Cross-domain Merging of Bayesian Networks

- **Merged network weight calculation**

$$w_{ij}^{\text{merged}} = \frac{\alpha_1 w'_{ij,1} + \alpha_2 w'_{ij,2}}{\alpha_1 + \alpha_2}$$

- **Cross-validated Log Likelihood (CVLL), MSE as evaluation metric**

- **CVLL**: Model fit under held-out data — higher is better

$$\ell_{k,j} = -\frac{1}{2}\left(n_{\text{te}}^{(k)} \log\left(2\pi\,\hat{\sigma}_{k,j}^2\right) + \frac{1}{\hat{\sigma}_{k,j}^2}\sum_{i\in\mathcal{D}_k}\left(y_{ij} - \hat{y}_{ij}\right)^2\right)$$

- **MSE**: Prediction error of the Gaussian SEM on observed data

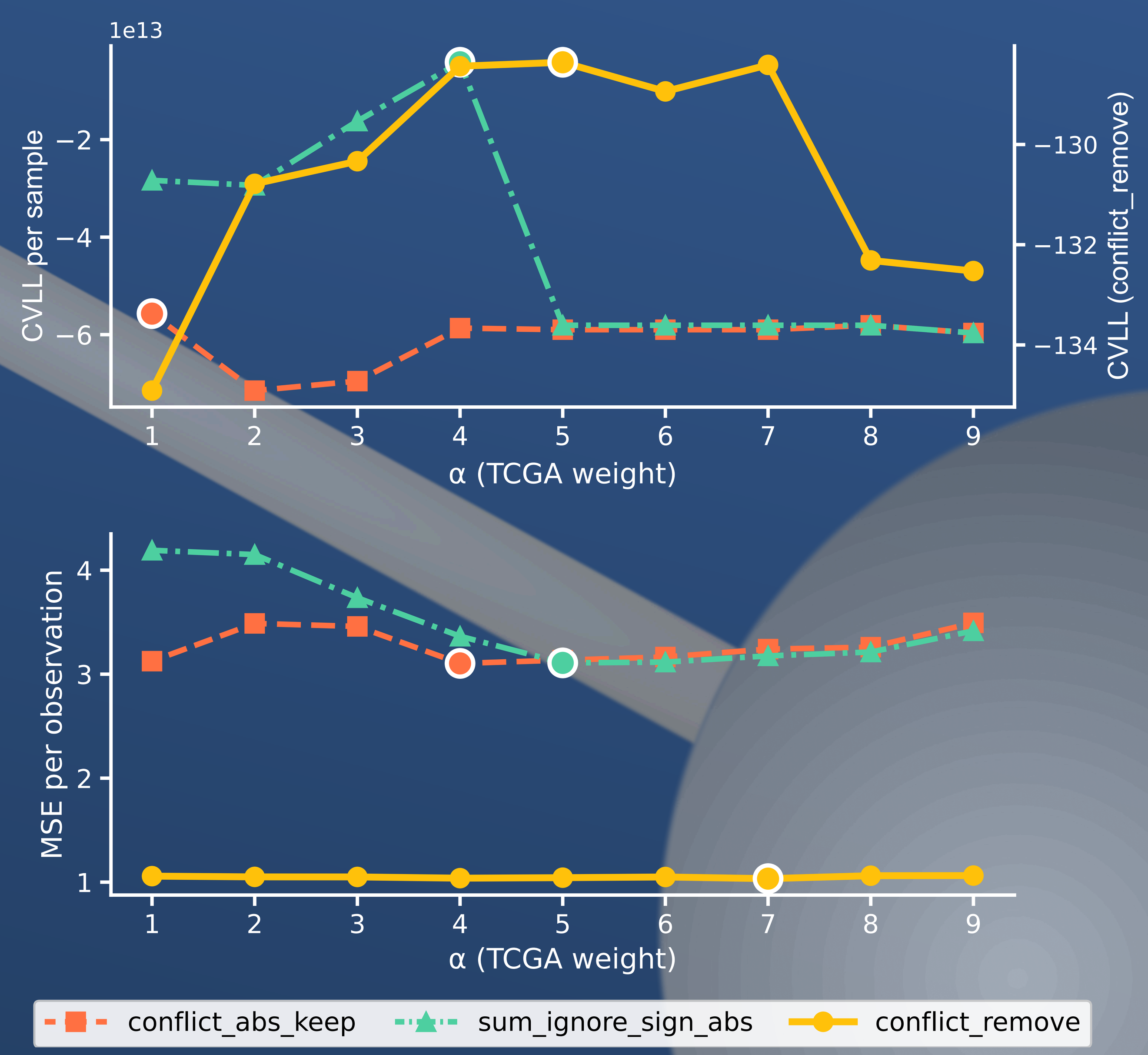
$$\text{SSE}_{\text{total}} = \sum_{k=1}^{K}\sum_{j=1}^{p}\sum_{i\in\mathcal{D}_k}\left(y_{ij} - \hat{y}_{ij}\right)^2 \quad \text{MSE}_{\text{per obs}} = \frac{\text{SSE}_{\text{total}}}{\sum_k n_{\text{te}}^{(k)} \cdot p}$$

Cross-domain Network Merging Methods Compared

■ **Three merge strategies compared (CVLL & MSE):**

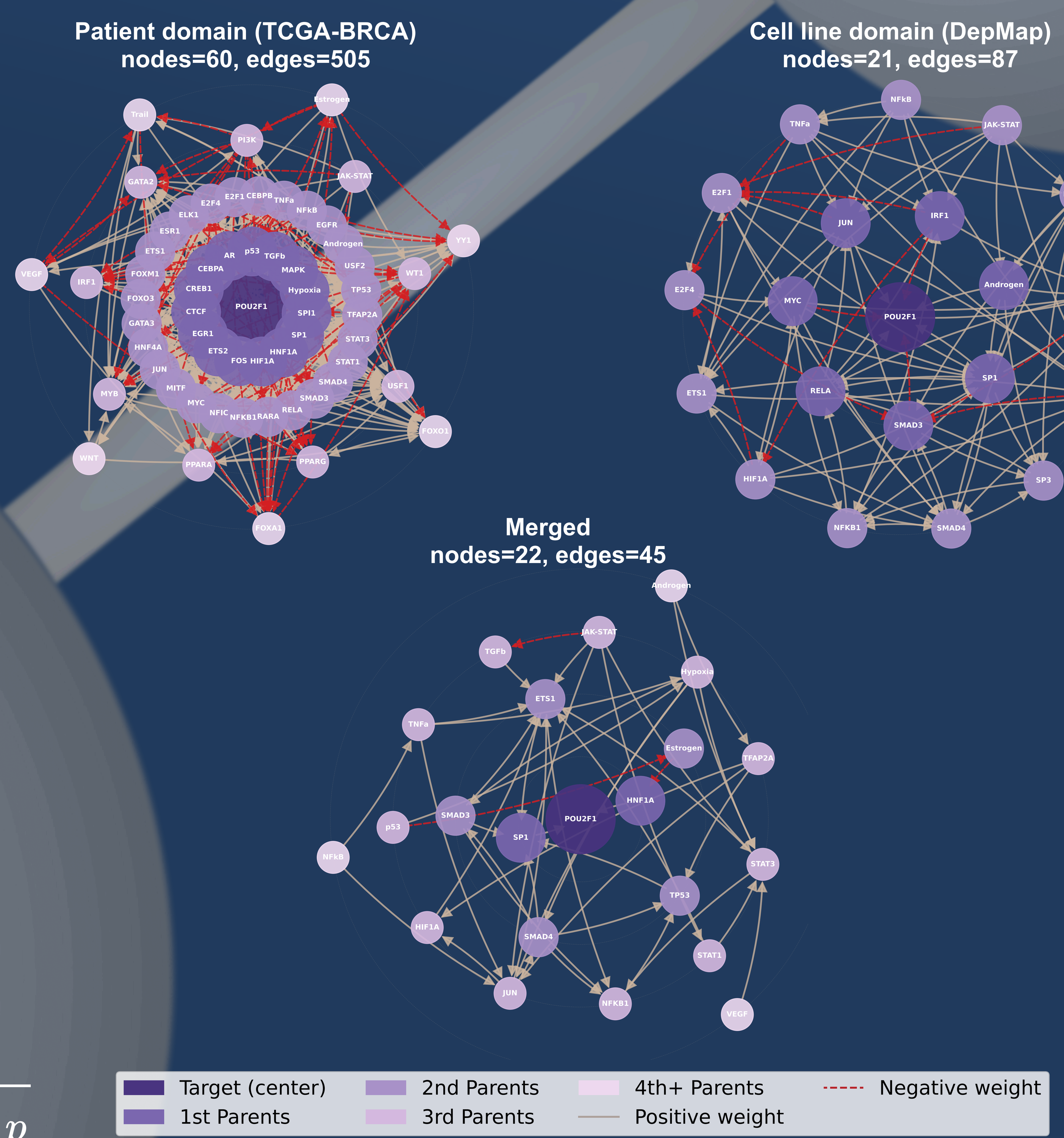
Strategy	Conflict handling
conflict_abs_keep	Keep edge, higher
sum_ignore_sign_abs	Keep edge, sum of
conflict_remove	***Remove conflicting edges***

Merge ratio sweep — all methods

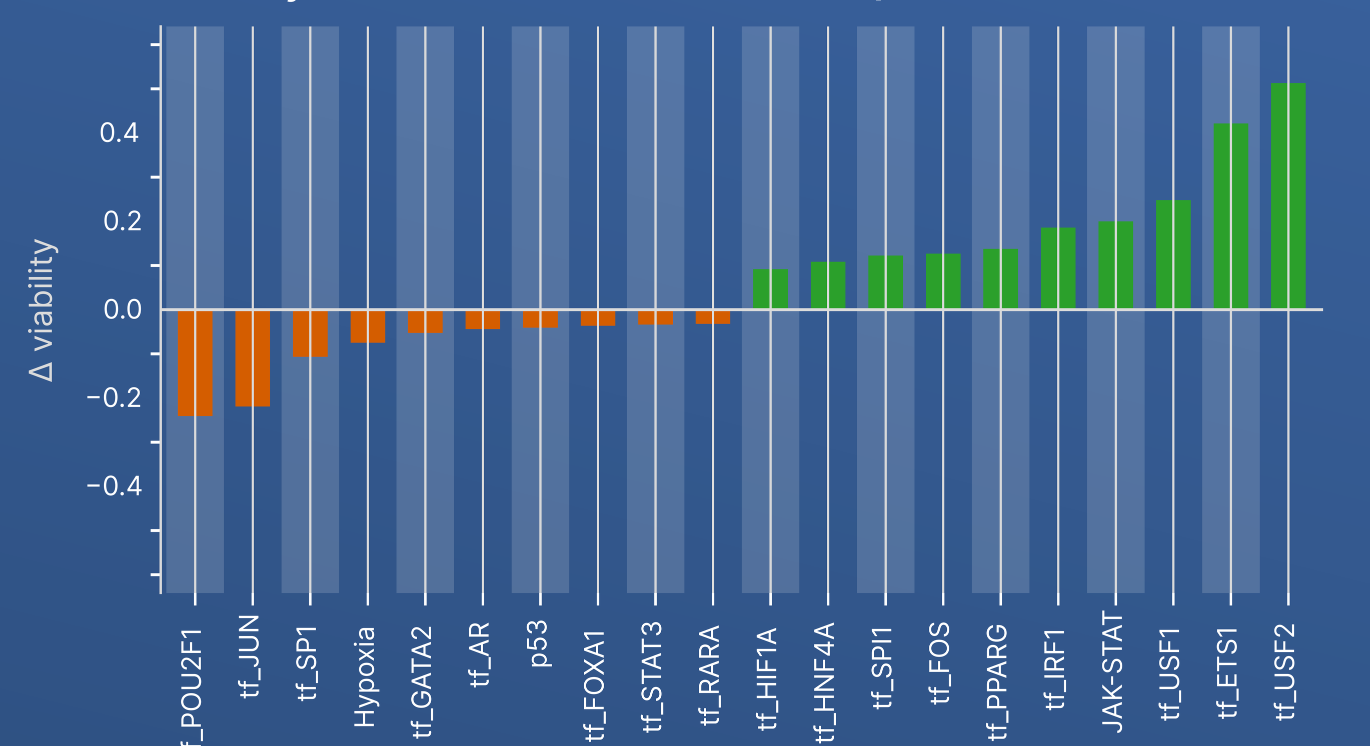

1e13
CVLL per sample
−2
−4
−6
−130
−132
−134
CVLL (conflict_remove)
1 2 3 4 5 6 7 8 9
α (TCGA weight)
MSE per observation
1
2
3
4
1 2 3 4 5 6 7 8 9
α (TCGA weight)
conflict_abs_keep
sum_ignore_sign_abs
conflict_remove

- **conflict_remove selected** — highest CVLL & lowest MSE across all α ratios

Merging Reveals New Connections Not Present in Cell Line


Patient domain (TCGA-BRCA)
nodes=60, edges=505
Cell line domain (DepMap)
nodes=21, edges=87
Merged
nodes=22, edges=45
Target (center)
1st Parents
2nd Parents
3rd Parents
4th+ Parents
Positive weight
Negative weight

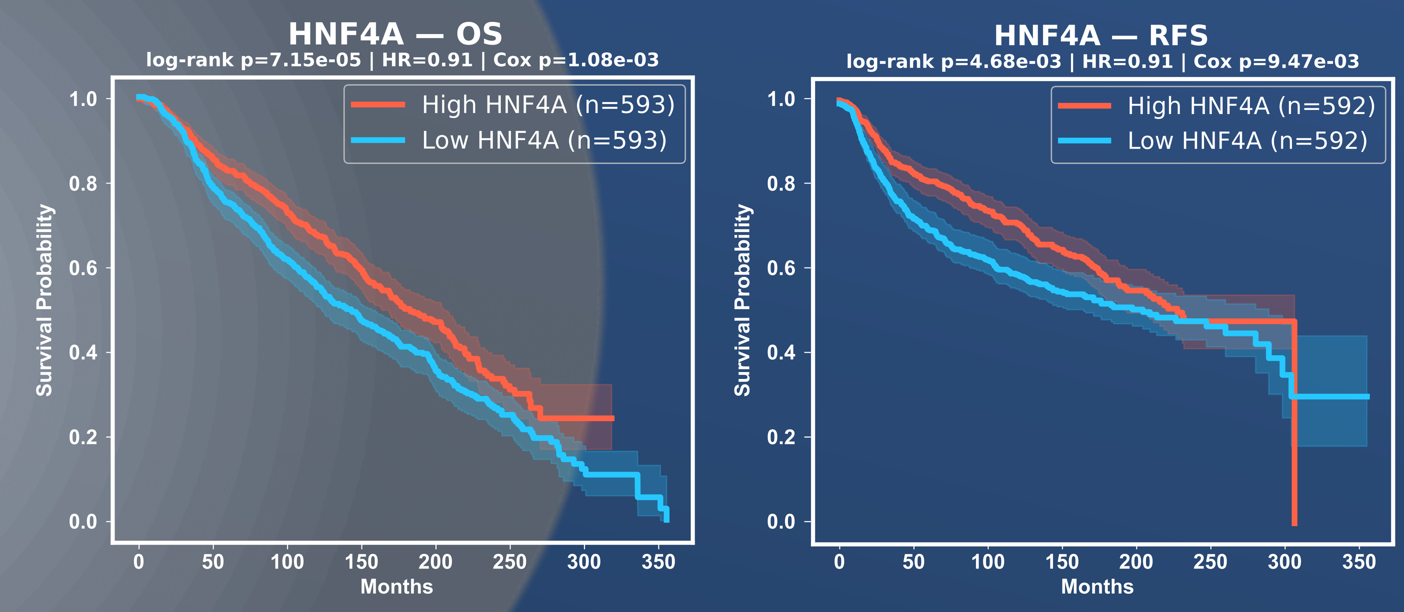
Do-simulation Identifies Causal TF Targets for Breast Cancer


Pathway/TF-level do-simulation score (-1σ)
Δ viability
0.4
0.2
0.0
−0.2
−0.4
tf_POU2F1
tf_JUN
tf_SP1
Hypoxia
tf_GATA2
tf_AR
p53
tf_FOXA1
tf_STAT3
tf_RARA
tf_HIF1A
tf_HNF4A
tf_SPI1
tf_FOS
tf_PPARG
tf_IRF1
JAK-STAT
tf_USF1
tf_ETS1
tf_USF2

- **Viability change after TF inhibition simulation (TF = μ−1σ)**
- **Orange (negative Δ):** TF inhibition → cancer cell viability ↓ — inhibition target (therapeutic candidate)
- **Green (positive Δ):** TF inhibition → cancer cell viability ↑ — TF suppresses cancer growth; loss accelerates tumor
- **Top inhibition candidates: tf_POU2F1, tf_JUN** (strongest red); tf_USF2, tf_ETS1 context-dependent (green)

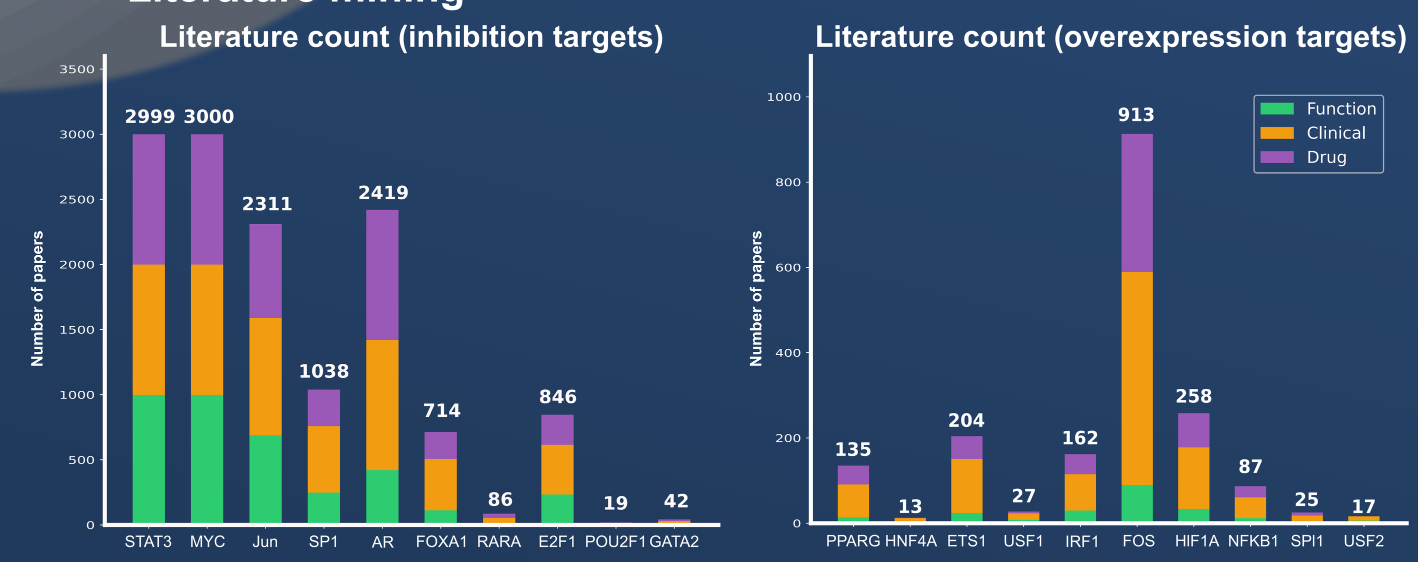
External Validation: Selected Targets Show Relevance

■ **METABRIC**


HNF4A — OS
log-rank p=7.15e-05 | HR=0.91 | Cox p=1.08e-03
High HNF4A (n=593)
Low HNF4A (n=593)
Survival Probability
Months
HNF4A — RFS
log-rank p=4.68e-03 | HR=0.91 | Cox p=9.47e-03
High HNF4A (n=592)
Low HNF4A (n=592)
Survival Probability
Months

- Patients grouped into High expression / Low expression based on DoRothEA TF activity
- Selected targets show clinical relevance

■ **Literature mining**


Literature count (inhibition targets)
Number of papers
2999 3000 2311 1038 2419 714 86 846 19 42
STAT3 MYC Jun SP1 AR FOXA1 RARA E2F1 POU2F1 GATA2
Literature count (overexpression targets)
Number of papers
135 13 204 27 162 913 258 87 25 17
PPARG HNF4A ETS1 USF1 IRF1 FOS HIF1A NFKB1 SPI1 USF2
Function
Clinical
Drug

- We mine “TFs” - Conventional pipeline focuses on “genes”
- Suggested targets include well known targets & novel targets

Conclusion

- BayesTX integrates patient and cell line transcriptomics with causal Bayesian Network framework, enabling interventional TF-level drug target prioritization and correlational analysis
- Cross-domain merge with bootstrap (B=200) yields 275 high-confidence causal edges, capturing both regulatory architecture and viability-linked signals
- Do-simulation identifies tf_POU2F1 and tf_JUN as top inhibition candidates
- Selected targets validated in independent cohorts: (METABRIC survival, GDSC2 drug sensitivity)
- Framework is cancer-type agnostic.

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